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Distributed Control of Wind Farm Power Set Points to Minimise Fatigue Loads

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Abstract— **The quantity and size of wind farms continue to grow as countries around the world strive to meet ambitious targets for renewable electricity generation such as the UK government's Net Zero target of increasing offshore wind energy from current levels (circa 6 GW) to circa 75 GW by 2050. With increasing size and quantity of wind farms, there is a growing requirement to use wind farm level control both to help with grid integration and to minimise the loads on the turbines in the farm. In this paper, a methodology of distributing power set points through a wind farm to minimise the loads on the turbines whilst meeting a delta power set point for the farm is presented. The methodology in this paper uses a hierarchical control structure, in which a network wind farm controller calculates the required change in wind farm power and then passes this value on to a distributed controller that defines the change in power required from each wind turbine. The network wind farm controller calculates a delta change in wind farm power that the wind farm holds in reserve. The distributed controller allocates the reductions in power output by first setting a baseline reduction that considers the steady state tower loads. The baseline is then adjusted to meet the required change in power, distributing the additional change in one of two ways; either proportional to the square of each turbines estimated wind speed or proportional to the initial baseline. Performance is assessed using the StrathFarm simulation tool. The wind turbine models incorporated into StrathFarm are sufficiently detailed to provide the tower and blade loads and the wind field model is sufficiently detailed to represent turbulence, wind shear, tower shadow and wakes and their interaction. The performance of the proposed wind farm controllers are assessed for a range of wind conditions for two 4x4 wind farms of 5MW wind turbines, one closely spaced (500m) and one less closely spaced (1000m). Both the accuracy of the change in power output from the wind farm and the change in turbines DELs are discussed. Depending on the wind conditions, the approach is found to reduce the tower and blade loads by about 10% more than the case in which each turbine is simply allocated the same change in power. There is good accuracy in the change in power at higher wind speeds. Below rated wind speed, wake effects reduce the accuracy of the change in power.**

I. Introduction

The trend in wind energy over recent years has been for wind turbine and wind farms to increase in size, especially in the offshore case. The increase in size of turbines and farms is driven by the increasing goals for renewable energy connection, of which wind is expected to play a large role given its relative maturity compared to other renewable technologies. As an example, the United Kingdom has laid out its "Net-Zero" plan for 2050, which aims to increase offshore wind energy from around 7GW today to around 75GW by 2050. Such a large increase in the volume of wind energy connected to the grid (note that the peak demand for the UK in 2018 was 50.3GW [1]) will necessitate that wind farms no longer operate in a "greedy" fashion as is often currently the case, whereby all wind turbines aim to maximise their own energy capture up to their rated power without consideration of wider network or farm considerations. Instead, it is likely that wind farms will be required to contribute in a more meaningful way to grid balancing, and turbines will need to consider the impact of their strategy on structural loading. Contributing to the stability of the wider grid can be done by holding power in reserve, sometimes called "delta control". Fast and accurate delta control allows wind turbines and/or wind farms to contribute to ancillary services such as synthetic inertia or droop control through variation of the power held in reserve. Reducing the power output of wind turbines can be done in many ways, with different advantages and disadvantages in terms of the speed of response to changes in power and the loads induced upon the turbine [3]. In this paper, a methodology is presented that allows the wind farm to supply accurate delta control, whilst distributing the required change in power of each turbine between the turbines in an intelligent way to minimise the loads. In section II the methodology and tools required to investigate the problem are detailed. The wind farm layout and environmental conditions for the simulations are detailed in section III. In section IV the results of the simulations are discussed, with conclusions drawn in section V.

II. Simulation Tools and Methodology

In order to simulate the proposed wind farm control approach, a suitable wind farm simulation tool is required. Simulation of large numbers of wind turbines can be highly computationally demanding for high fidelity simulations such as those using blade element momentum theory (BEM) or CFD, however, running low fidelity simulations (such a simple power curve look up or a basic actuator disc model) does not provide a good enough assessment of the dynamics of the turbines and the loads on the turbines in order to accurately assess the loads resulting from the control action and site conditions simulated. In order to bridge the gap and

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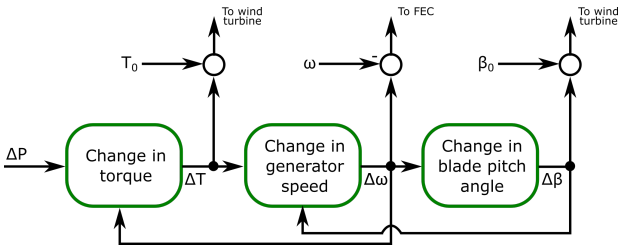


Fig. 1. A simplified flow chart of the PAC algorithm. The flow chart is a useful way to think through the operation of the PAC, with the change in power, ΔP, leading to a change in torque ΔT, which in turn is used to find a change in generator speed $\Delta\omega$, which drives the calculation of the change in pitch angle $\Delta\beta$.

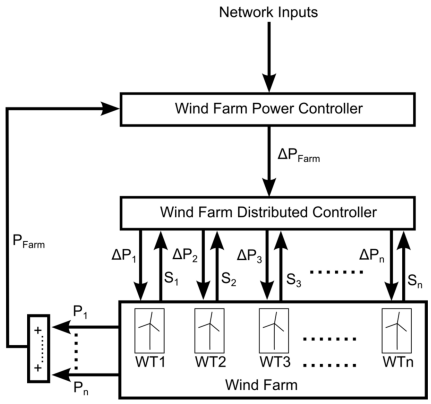


Fig. 2. Hierarchical Control Structure

allow multiple simulations to be run whilst still providing the necessary level of complexity to assess individual turbine dynamics and the relative loads induced through different control approaches, the medium fidelity "StrathFarm" tool has been developed. StrathFarm uses wind turbine models, developed over the past 20 years, which incorporate lumped parameter models of the drive-train and rotor, aerodynamic modelling that includes induction lag effects, and modelling of blade and tower loads validated against DNV GL's Bladed. The wind comprises a pre-processed correlated low frequency wind farm wind field, to which higher frequency effective wind field components are added locally to each turbine. Wakes can be modelled in two ways; an engineering model based on the work of Frandsen [4] or a higher fidelity method based on the WFSim model [2]. Modelling of wake meandering and partial wakes is included. In this work the Frandsen method is used. StrathFarm is uniquely placed in providing a simulation environment that allows medium fidelity simulations modelling turbine dynamics and wakes in turbulent winds to be run in better than real time for wind farms of up to 100 wind turbines. More detail of the StrathFarm simulation tool can be found in [5]. The turbine used is the SuperGen 5MW wind turbine, developed as part of the SuperGen project and based on the NREL 5MW exemplar turbine. [7].

A. Wind Farm Control (WFC)

Strathfarm is designed as tool for developing wind farm controllers and so each turbine is equipped with a Power Adjusting Controller (PAC) [8] that allows the operating point of the wind turbine to be altered without any changes being made to the full envelope controller, i.e. the PAC is completely independent of the full envelope controller design. Additionally, the performance of the full envelope controller is not compromised by the presence of the PAC, with minimal change to the feed-forward gain. A simplified but useful overview of the PAC's operation is shown in Fig. 1. The requested change in power ΔP is implemented through a change in the generator torque ΔT, which results in a change in the generator speed $\Delta\omega$, which is in turn ameliorated (if possible) by a change in the blade pitch angle $\Delta\beta$. By effecting the change in power through the torque demand the PAC is able to deliver fast and accurate changes in power output, whilst a slower control loop minimising the error in generator speed ensures that the rotor speed is kept close to the desired value. The PAC was designed specifically to allow complete flexibility in operation, i.e. to allow provision of any demanded change of power that is possible for the turbine to safely provide. The PAC has safety features that prevent the wind turbine's operating point from moving outside of a defined safe working envelope [8]. If the operating point moves outside of the safe working envelope then the PAC will limit its output and in some cases enter a recovery mode where the turbine is returned back to its usual operating point. When in recovery mode the PAC does not deliver the requested ΔP as changes to the torque and pitch are required to return to normal operation. The WFC is designed in a hierarchical manner as shown in Fig. 2, whereby network inputs (i.e. inputs from outside of the wind farm controller from say, the grid operator) are processed by the wind farm power controller to calculate the desired change in the wind farm's power output ΔP_{Farm}. The change in wind farm power output is input to the wind farm distributed controller, which decides how the change in power should be distributed amongst the wind turbines in the farm via the individual turbine changes in power $\Delta P_1, \Delta P_2, \ldots, \Delta P_n$. The turbine change in power is applied via the PACs on each turbine, which also feedback signals $S_1, S_2, \ldots, S_n$. The total power output of the farm is fed back to the wind farm power controller. It is important that the change in power for an individual turbine is derived from the power output of a farm of turbines (at least 4 others) in order to prevent a strong feedback loop that would negatively impact on the operation of the wind turbine's full envelope controller. The hierarchical method used means that the WFC is able to be as flexible as possible, i.e. the WFC is minimally constrained in its functionality. By separating the wind farm power controller and the wind farm distributed controller, the two components can be designed separately and a "mix and match" approach can be used to ensure the particular goals of a particular wind farm controller are met by combining an appropriate power controller with an appropriate distributed controller.

1) Wind Farm Power Controller: In this paper a "delta control" is applied. In order to calculate the required change in power an estimate of the farm power without the WFC being applied (P_0) is required, which is the sum of the turbine power without the PAC, calculated in the PAC by using a power curve look up table with the estimated wind speed at each turbine (including induction lag effects) as the input. Hence, the desired farm power P_{des} is calculated by,

$$P_{des} = KP_0 \tag{1}$$

where K is a number less than 1 and greater than 0. An error, ϵ, can hence be calculated between the desired farm power and the measured farm power.

$$\epsilon = P_{des} - P_{Farm} \tag{2}$$

A PI controller is then used to minimise the error. The proportional and integral gains are found heuristically and have the values 0.8 and 0.05 respectively.

2) Wind Farm Distributed Controller: The wind farm distributed controller, referred to hereafter as simply the distributed controller, is designed to distribute the individual turbine changes in power in a manner that,

- Produces the required change in wind farm power ΔP_{Farm}
- Achieves some other auxiliary goal

For the work presented here, the auxiliary goal is to minimise the fatigue loads on the wind turbines in the farm. Fatigue loads are dependent on the magnitude of the bending moment, the number of load cycles, and the variation in the moment over the cycle. In this paper damage equivalent loads (DELs) are calculated in the time domain using rainflow counting to determine the stress cycles and then these are converted to damage using Wohler SN-curves and Palmgren-Miners rule (calculated in this paper using the approach of [6] itself based on IEC standards). As discussed in section II-A, the PAC effects the requested change in power through torque action, with pitch actuation used to ameliorate any induced change in rotor speed. The pitching of the blades results in a change to the fore-aft tower moment and the blade edge and flap moment. Whilst the loads on a wind turbine's blades and tower are dynamically changing and therefore, a methodology that takes into account the measured loads when distributing the turbine changes in power has some merit, the loads on the blades and tower are not always measured on a wind turbine. Hence, whilst direct measurements of the loads could be used, a simpler methodology, based on the static loads on the turbine is applied here such that it could be applied to any wind turbine.

An approach that utilises the static loads has potential to produce good reductions in fatigue loads, as cycles of loading at higher mean stress are more damaging than similar cycles at lower stresses. Additionally, reducing the moments more around the peak "flattens" the curve, reducing both the peak moment and the gradient of the moments. As such, low frequency cycles that more or less follow the curve will be less damaging as the amplitude of the variation and the mean stress will be lower.

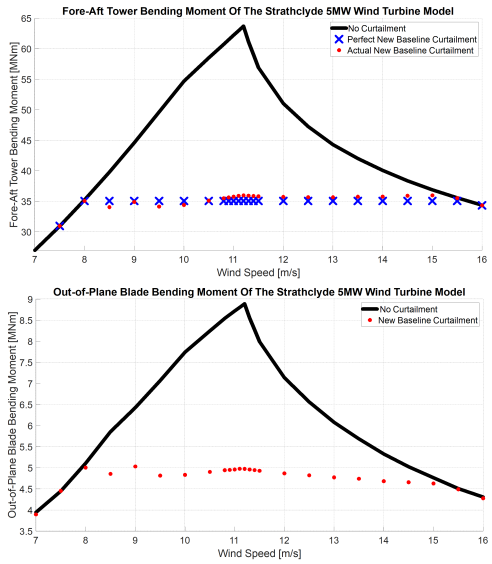


Fig. 3. Static fore-aft bending moments on the wind turbine tower (top) and static out of plane bending moments on the wind turbine blades (bottom)

The static out of plane loads on the wind turbine blades and fore-aft loads on the tower are shown in Fig. 3. It can be seen that both out-of-plane blade bending moments and tower fore-aft bending moments increase from a minimum at low wind speeds towards a peak at the rated wind speed (just over 11 m/s), before reducing again as the wind speed continues to increase. The reduced moments in low wind conditions are due to lower thrust from the wind, whilst the reduced moments in higher wind speeds are due to the wind turbine blades pitching to feather and hence reducing the thrust coefficient of the rotor. It is clear from Fig. 3 that pitching the blades can have a significant effect on the structural bending moments and it is true that pitching the blades due to a power reduction via the PAC results in a similar change in thrust force, and hence could potentially reduce the blade out-of-plane and tower fore-aft loads. Given the information shown in Fig. 3 it is hypothesised that allocating the change in power in a manner that evens the tower loads over the turbines in the wind farm could lead to an overall reduction in the fatigue loads. The hypothesis is based on a reduced variation in the structural loads over low frequencies and a lower mean load due to the peak of the bending moment shown by the black solid line in Fig. 3 (upper) being flattened as shown by the blue crosses. The strategy designed to achieve this aim is for wind turbines in the farm to follow the contour plot shown in Fig. 4 as a baseline strategy where the baseline change in power ΔP_{nBase} is dependent upon the wind speed as estimated by the turbine's PAC. The slight difference between the exact horizontal line of perfectly flat tower bending moment in Fig. 3 (blue crosses) and the actual baseline followed (red dots) is that a linear approximation to the curve in Fig. 4 (shown as a red line) is used. As the blade moments are similarly flattened it is hypothesised that the fatigue loads in the blades

will be similarly reduced. Note that because the wind speed estimate is highly decoupled from the turbine dynamics there is no strong feedback loop induced around the turbine (for much greater detail on the wind speed estimator and PAC see [8]). The wind speed is low pass filtered and passed through a look up table to calculate the desired value of ΔP_{nBase}. Clearly, following the baseline on its own will not necessarily result in the correct change in power and so the required additional change in power ΔP_{nAdd}, which may be positive or negative, is included for each turbine such that, for the nth wind turbine in a farm of N available turbines,

$$\Delta P_n = \Delta P_{nBase} + \Delta P_{nAdd} \tag{3}$$

Initially, the required additional change in power ΔP_{add} is allocated to the wind turbines proportionally to the estimated wind speed from each turbine's PAC squared ($\hat{V}_n^2$). This approach was taken with a focus on the reduction in tower moment at higher wind speeds. Hence the total change in power for the nth wind turbine in a farm of N available turbines,

$$\Delta P_n = \Delta P_{nBase} + \frac{(\Delta P_{Farm} - \sum_{n=1}^{N} \Delta P_{nBase})\hat{V}_n^2}{\sum_{n=1}^{N} \hat{V}_n^2} \tag{4}$$

A second approach was also considered, in which the initial change in power was again distributed as per P_{nBase}. The remaining change in power required to achieve the desired change in power output (ΔP_{nAdd}) is then distributing proportional to P_{nBase}. In effect, this approach raises or lowers the tracked curtailment curve (red line in Fig. 4) such that the correct curtailment is achieved. Algebraically this approach is described by,

$$\Delta P_n = \Delta P_{nBase} + \frac{\Delta P_{Farm} - \sum_{n=1}^{N} \Delta P_{nBase}}{N} \tag{5}$$

For comparison, a third strategy is also simulated, whereby the change in the farm power is simply evenly spread amongst the available turbines in the farm. This first methodology is referred to herein as the Intelligent Distributed Controller (IDC), the second methodology is referred to herein as the Intelligent Distributed Controller 2 or (IDC2), whilst the third is referred to as the Evenly Distributed Controller (EDC).

Because the controller contains integral action, it is necessary to include anti-wind up to prevent integral wind up when the limits of the actuator (in this case the PAC and turbine) are exceeded. The back calculation method described in [8] is used for this purpose. The rate of change of power is also limited to prevent unwanted loads on the turbine from sudden changes in torque. The same limits are applied for all control approaches used in this work. Note that only "available" turbines are considered for the distributed controller. An available turbine is one that,

- Has an effective wind speed estimate above the minimum wind speed (in this case 6m/s)
- Is in a safe part of the operational envelope as determined by the signals from the PAC [8]
- Is not currently undergoing the recovery process to return to normal operation

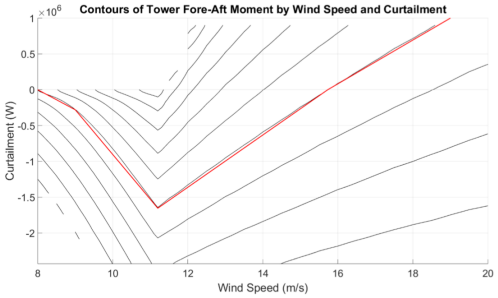


Fig. 4. A plot of contours of constant tower fore-aft moment by curtailment and wind speed with the proposed baseline curtailment strategy

III. Wind Farm Layout and Environmental Conditions

The simulation results presented in this paper use a wind farm of 16 SuperGen 5MW wind turbines arranged in a 4x4 square. Whilst simple, such an arrangement is common for offshore wind turbines. Simulations are conducted for all permutations of the following conditions:

- Mean wind speeds 9m/s, 12m/s, 15m/s, and 18m/s
- Wind directions 0° (perpendicular to the side of the square), 22.5^o, and 45° (note that due to the symmetric arrangement of the wind farm these directions are also valid for their value plus 90°, plus 180°, and plus 270°).
- Turbulence levels of 10%
- Power reduction of 10% of normal farm power output at the wind speeds experienced by the farm.
- Three WFC designs (IDC, IDC2 and EDC as described in section II-A) and the base case where no WFC is used.

Simulations are conducted for a duration of 1,000 simulation seconds. The initial 400 seconds were then discarded to eliminate any transient effects of the simulation start-up. Two wind farm layouts were investigated. One in which the spacing between turbines was typical of current wind farms (8 diameters, 1,000 metres) and one in which the turbines were more closely spaced (4 diameters, 500 metres).

IV. Results

For each simulation conducted the two specific aims of the controller as stated in section II-A.2 must be considered, specifically,

- What was the effect on the tower fore-aft and blade out-of-plane fatigue loads
- Was the delta control accurate?

To assess the tower fore-aft and blade out-of-plane fatigue loads the damage equivalent loads (DELs) for each case are calculated. The average fatigue loads of the turbines in the farm for each wind speed and direction combination are shown in Table I for the 500m separation and in Table

II for the 1000m separation. The DELs are shown as the percentage difference from the non-WFC case. All the percentage changes for IDC and IDC2 are negative, so in all cases the DELs were decreased through the control action. In all cases the IDC and IDC2 control methods outperformed the EDC controller, except at 12m/s for the 500m separation in which EDC outperforms IDC for all directions at 500m separation and at an angle of 22.5 for 1000m separation. This may be due to the squared wind speed component of the distributed controller that skews the reduced power towards wind turbines experiencing higher wind speeds, which, in this case, are actually less loaded. IDC2 results in similar performance to EDC in this case. For wind speeds of 9m/s, 15m/s and 18m/s the IDC and IDC2 controllers outperform the EDC controller in both tower and blade DEL reduction by as much as 20% for tower loads and 27% for blade loads, with a reduction of 10% in DELs compared to EDC typical.

An average reduction in DELs is a good result, however, it is important that the average reduction is not a result of large reductions for a small number of turbines and increases in DELs for others. It is preferable that all turbine DELs are reduced. The plot in Fig. 5 shows the change in DELs compared to the base case of no WFC for each turbine within the farm and for each control strategy used for all the 1000m separation simulations. It is clear that alomost all turbines experience a reduction in DELs for IDC and IDC2.

To assess the accuracy of the delta control the power output of the controlled cases are compared to 90% of the power of the non-controlled case (perfect delta control) and the mean error is calculated. The results are shown at the bottom of Tables I and II. The mean percentage error in the change in power is high for low wind speeds and independent of control type, whilst at high wind speeds the error is much smaller but with some variation correlating with control type. The large errors at low wind speeds are because the change in farm power ΔP_{Farm} does not take into account the changes in the wakes of the wind turbines caused by altered operation. The error is largest in the case of the 9m/s average wind speed simulations, as in below rated wind conditions a reduction in the effective wind speed has a cubic relationship to the reduction in power. In simulations well above rated the wakes have a minimal effect on power as the turbine is already limiting its power to the rated value through pitch actuation and hence the mean error is much smaller. The effect is highlighted in Fig. 6, which shows plots from the simulation at 9m/s and 0 degrees wind with 500m separation. The upper plot in Fig. 6 shows the farm power output without wind farm control, the ”perfect” total farm output of 90% of farm power output without wind farm control and the power output with IDC and EDC. It is clear that both IDC and EDC have substantial errors. In the middle plot the estimate of power without WFC is shown to be accurate when no WFC is in use. When using IDC or EDC however, the estimate is no longer accurate. The average wind speed estimate across the farm is shown to be increased in the lower plot of Fig. 6. The clear explanation is that the reduction in power of upstream wind turbines has increased

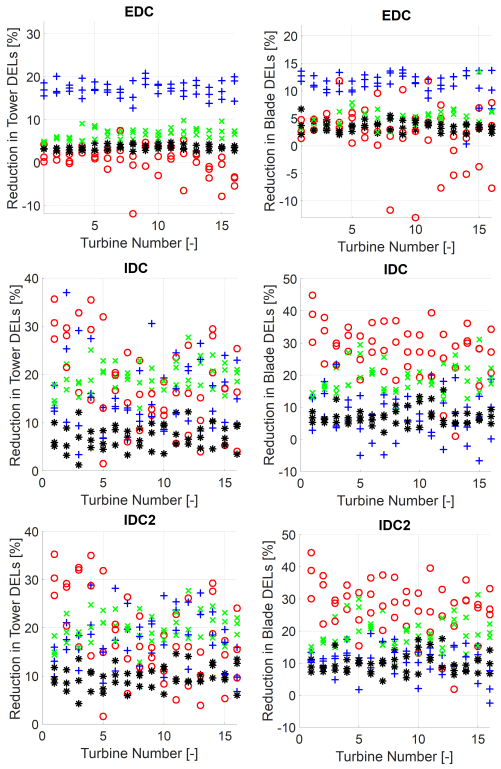


Fig. 5. Tower DEL (left column) and Blade DEL (right column) reductions for each turbine for all 1000m separation simulations, red o 9m/s, blue + 12m/s, green x 15m/s, black * 18m/s

the wind speed in downstream turbines, leading to a change in the wind speed experienced by the downwind turbines and hence an increase in the wind speed estimate, leading to an over-estimate of the power output without WFC in use. For high wind speeds, IDC2, and to a lesser extent IDC require greater rates of change of power, in some cases activating the rate limits of the WFC and resulting in errors. As the same turbulence of 10% is used for all simulations the magnitude of wind speed changes due to turbulence and hence power rates of change are higher at higher wind speeds.

V. Conclusions

A wind farm controller designed to minimise the structural loads on the wind turbines in the farm whilst providing a ”delta” change in power of 10% of the non controlled output was designed and tested using the StrathFarm wind farm simulation package. At lower wind speeds the power tracking is inaccurate, with an error of up to 37% and 16% for 500m and 1000m spacing respectively. The poor performance in low wind speeds is due to the diminished wakes of the upstream turbines increasing the power output of the down stream turbines. Tracking is better at higher wind speeds but rate limits introduce some errors for IDC and IDC2. Future work should focus on how to remedy this error.

The IDC and IDC2 strategies showed excellent reductions in tower and blade DELs. Both showed similar reductions however IDC2 is more consistent. Reductions of between 3% and 23% compared to no power reduction were observed

Mean Wind Speed [m/s]	9			12			15			18		
Direction [deg]	0	22.5	45	0	22.5	45	0	22.5	45	0	22.5	45
IDC Change in Tower DELs [%]	-12	-12	-3	-10	-14	-14	-22	-23	-22	-8	-8	-10
IDC2 Change in Tower DELs [%]	-12	-11	-8	-16	-18	-19	-22	-23	-22	-10	-11	-13
EDC Change in Tower DELs [%]	-1	2	1	-14	-17	-17	-10	-10	-10	-4	-4	-5
IDC Change in Blade DELs [%]	-26	-26	-23	-10	-14	-14	-23	-23	-23	-7	-8	-8
IDC2 Change in Blade DELs [%]	-27	-25	-21	-10	-12	-12	-22	-23	-23	-10	-11	-11
EDC Change in Blade DELs [%]	-6	1	5	-9	-13	-13	-7	-7	-7	-3	-4	-4
IDC Mean Error [%]	26	34	36	19	4	10	5	3	4	5	5	14
IDC2 Mean Error [%]	25	33	37	21	6	13	4	4	5	3	2	7
EDC Mean Error [%]	30	37	37	19	5	11	-1	-1	-1	-2	-1	-1

TABLE I

DELS FOR EACH STRATEGY COMPARED TO NO REDUCTION IN POWER AND MEAN ERROR FOR EACH STRATEGY - 500M SEPARATION

Mean Wind Speed [m/s]	9			12			15			18		
Direction [deg]	0	22.5	45	0	22.5	45	0	22.5	45	0	22.5	45
IDC Change in Tower DELs [%]	-19	-20	-17	-17	-12	-19	-19	-21	-18	-10	-7	-6
IDC2 Change in Tower DELs [%]	-19	-19	-18	-18	-17	-18	-20	-21	-19	-12	-10	-8
EDC Change in Tower DELs [%]	-1	0	-1	-17	-17	-18	-7	-7	-6	-4	-3	-3
IDC Change in Blade DELs [%]	-29	-30	-25	-10	-3	-10	-19	-20	-17	-9	-7	-6
IDC2 Change in Blade DELs [%]	-28	-30	-25	-11	-11	-11	-19	-20	-18	-11	-9	-9
EDC Change in Blade DELs [%]	-2	-3	-2	-12	-10	-12	-5	-6	-5	-4	-3	-4
IDC Mean Error [%]	16	12	11	0	5	0	0	-1	0	6	5	3
IDC2 Mean Error [%]	16	12	10	1	6	1	-2	-3	-2	2	3	1
EDC Mean Error [%]	16	12	14	1	5	1	-1	-1	-1	1	2	2

TABLE II

DELS FOR EACH STRATEGY COMPARED TO NO REDUCTION IN POWER AND MEAN ERROR FOR EACH STRATEGY - 1000M SEPARATION

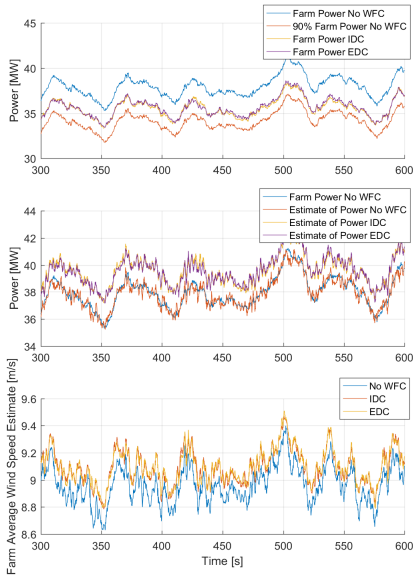


Fig. 6. Measured farm power, estimated farm power and estimated wind speed with different controllers, closely spaced wind farm

for 500m separation and between 12% and 21% for 1000m separation. The reduction in tower DELs for IDC2 was always greater than the more simplistic EDC strategy that simply divided the required change in power equally between available turbines. The principle of distributing the changes in turbine power based on the thrust curve and the wind speed is shown to be very promising for reducing loads. Possible future work to build on these results includes, methods for accounting for wake effects in calculating the desired change in farm power, investigation of different wind farm layouts, a greater number of simulations to minimise uncertainty and consideration of how measures/estimates of the dynamic loading on the blades and tower could be utilised to further reduce fatigue loads.

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